CAPSTONE PROJECT

Global Economic Analysis

Growth and the green transition 2000–2023

United States
China
Russia
India
Canada
Australia

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The economic growth challenge

Can major economies expand while reducing their carbon footprint?

The problem

The project examines whether clean energy adoption keeps pace with industrial energy demand.

Why it matters

Persistent fossil fuel dependence can increase climate exposure and threaten long-term economic resilience.

The risk

If we don’t separate economic growth from pollution, climate change will get worse, leading to more extreme weather, rising costs and damage to the global economy

A Quick Metaphor

Think of an economy as a car.

As the country grows, the car goes faster and needs more fuel.

Instead of switching the engine to run on electricity (renewable energy), the car is still running almost entirely on gas (fossil fuels).

The faster the car goes, the more exhaust fumes (carbon emissions) it pumps into the air.



Global Economic Indicators Analytics

DATA STRUCTURE

One record per country and year

A data dictionary defining indicators and their units across multiple years.

Analytical uses

Country benchmarking, trend analysis, forecasting and policy or sustainability studies.

INDICATOR COVERAGE

Economy and population

GDP, inflation, labor markets and demographics

Health and social development

Healthcare, education, poverty, equality and HDI

Energy, environment and infrastructure

Energy, renewables, CO2, land and infrastructure

Governance, security and tourism

Corruption, press freedom, voting, crime, military and tourism

Source: supplied data dictionary. The file defines the dataset but provides no observed values or analytical findings.

Research questions and objectives

01 GDP and CO2 emissions

Do economies show decoupling over time?

02 The energy mix

How does renewable adoption evolve relative to total energy use?

03 Environmental footprint

How does industrial activity relate to resource consumption?

DESCRIPTIVE OBJECTIVES

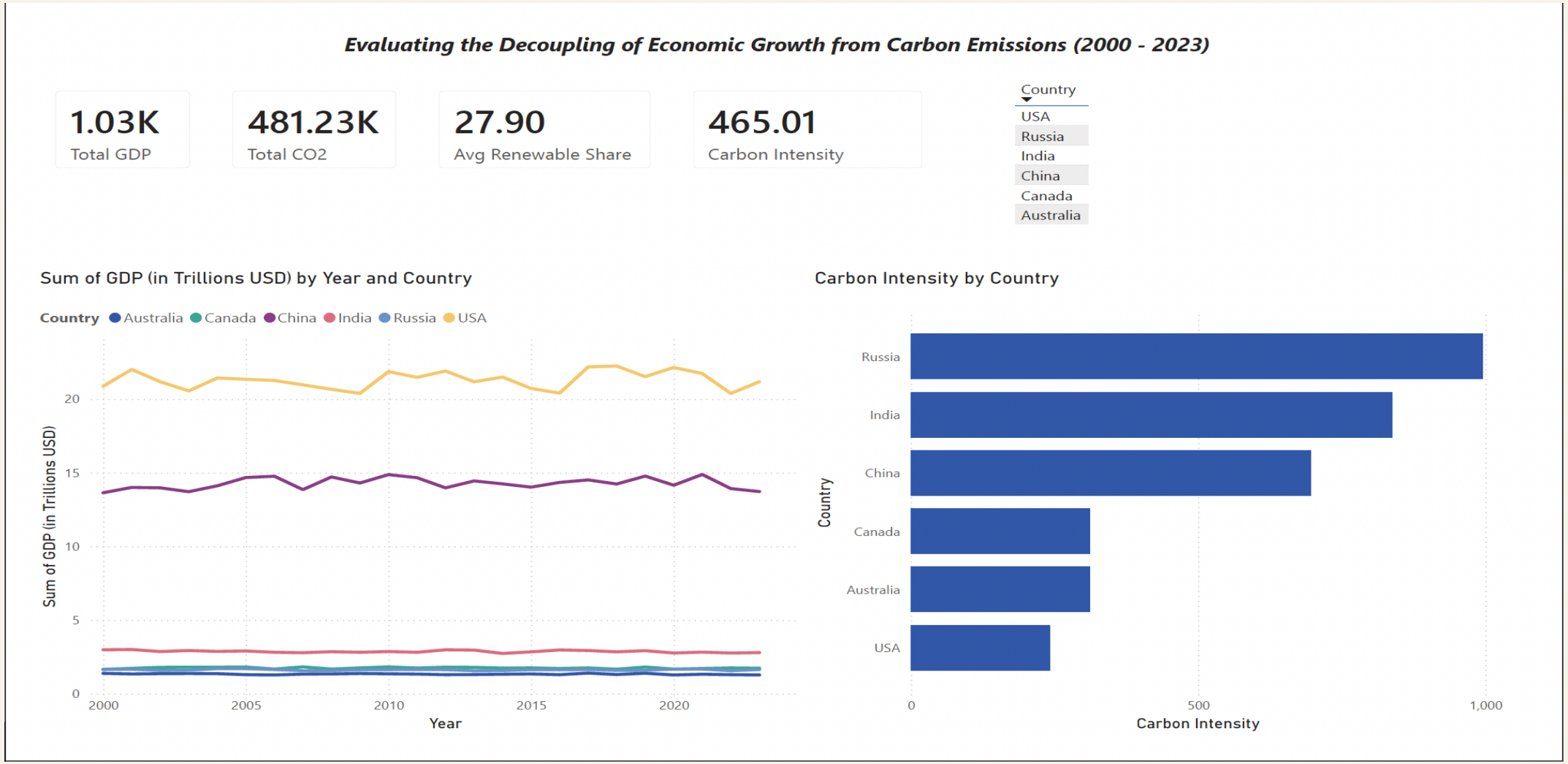
Compare historical trends and country energy dependencies. Examine resource impacts.

PREDICTIVE OBJECTIVES

Explore an Environmental Kuznets Curve model and renewable energy forecasts to 2030.

01 GDP and CO2 emissions

Do economies show decoupling over time?



Line chart: Visually proves whether GDP growth is scaling faster than emissions.

Bar chart: show which economies are cleaner per dollar of GDP.

Growth and emissions can follow different paths

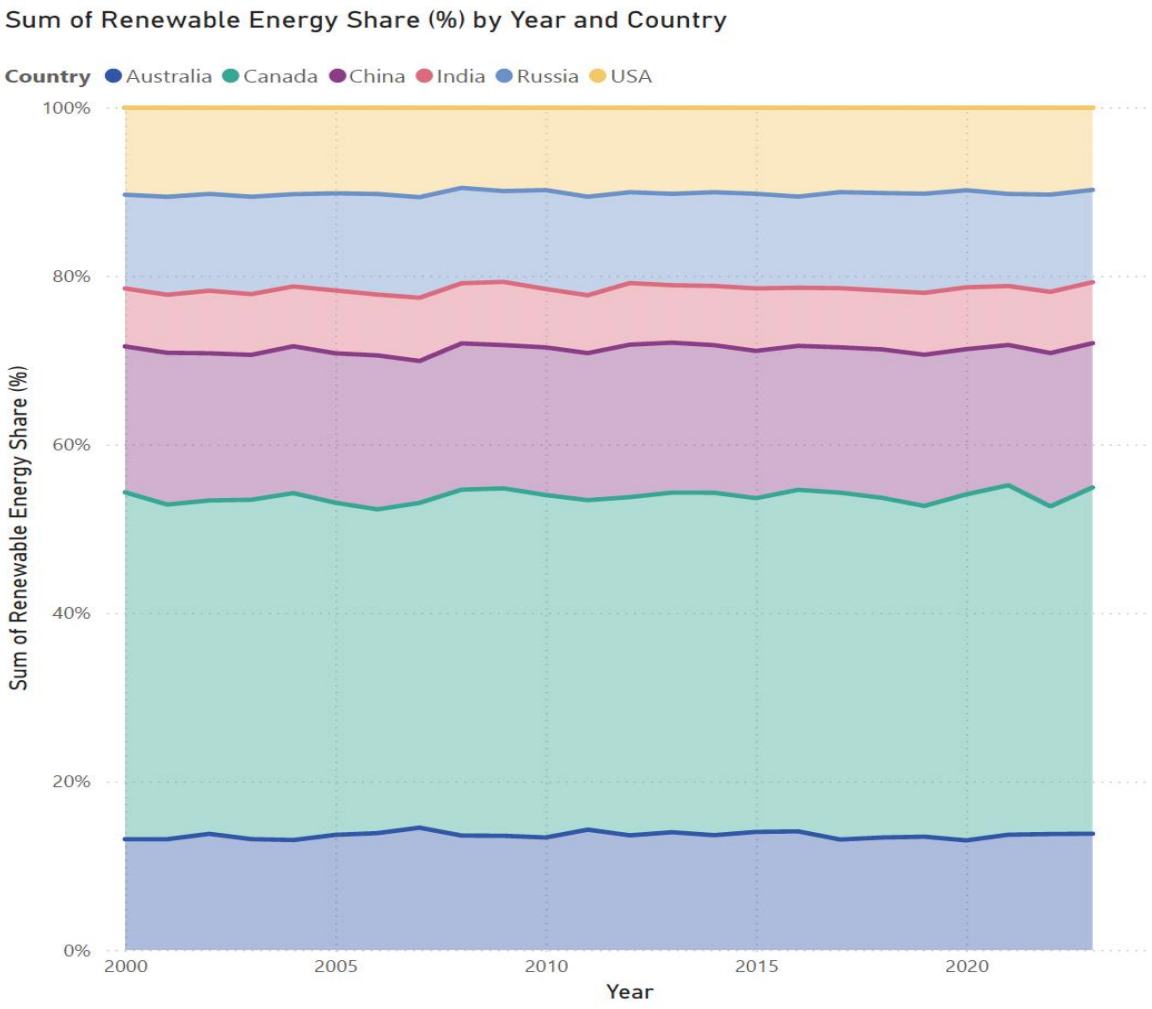
Reported finding: GDP growth does not always coincide with a proportional rise in CO2 emissions.

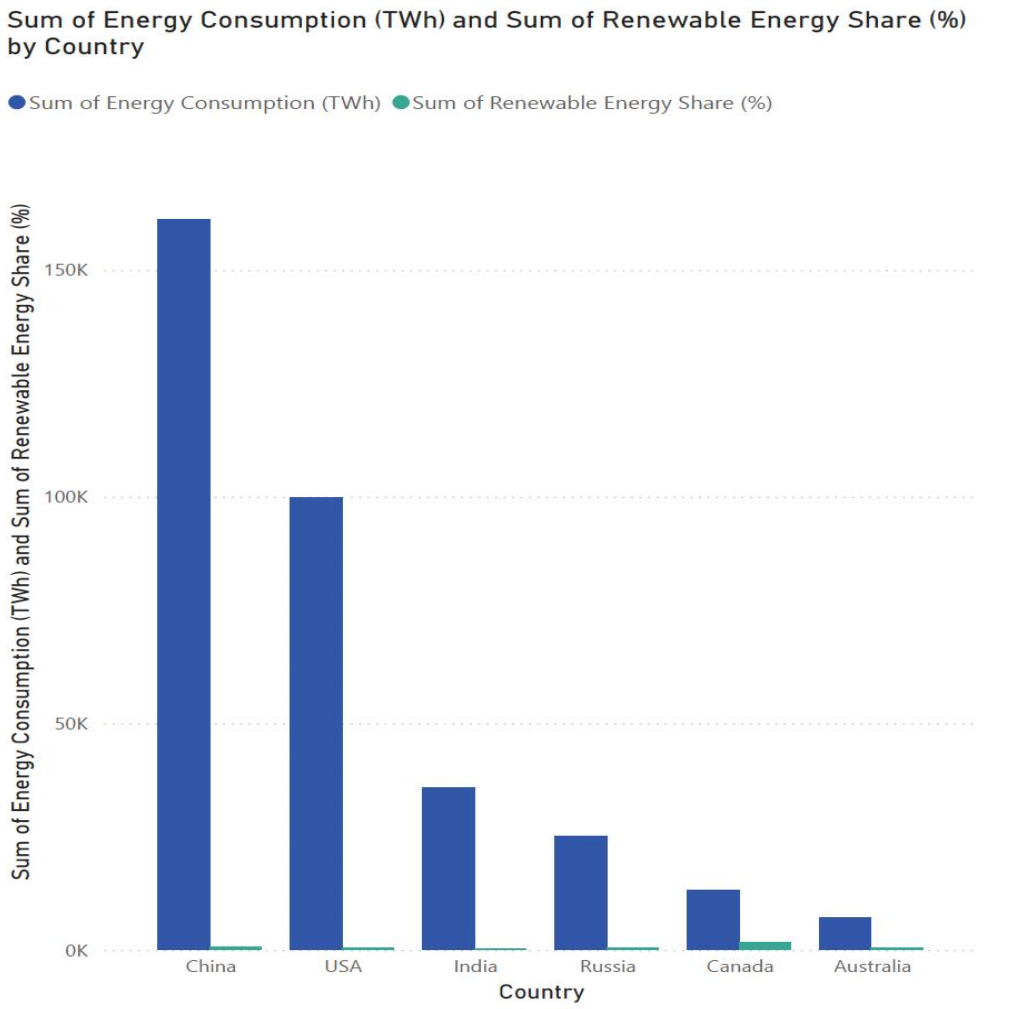
Decoupling concept	Pattern during economic growth
Relative decoupling	Emissions grow more slowly than economic output
Absolute decoupling	Economic output grows while total emissions fall

Evidence limit The summary does not establish which countries meet either test. Statistical nonsignificance alone cannot demonstrate structural decoupling.

02 The energy mix

How does renewable adoption evolve relative to total energy use?





Contrasts developed economies (like Canada and Australia) against emerging powerhouses (like China and India) regarding renewable adoption rates.

Energy Consumption (TWh) vs. Renewable Energy Share (%) to highlight structural dependencies on fossil fuels.

Renewable adoption is increasing unevenly



No country ranking or numerical growth rate accompanies the supplied summary.

Reported finding

Renewable energy adoption is rising, with uneven progress across the economies studied.

Comparison lens

Compare the renewable share with total energy use to assess whether clean supply keeps pace with demand.

Industrial activity and environmental pressure

Reported finding

Industrial expansion and fossil fuel dependence continue to put pressure on environmental sustainability.

Useful derived measures

Carbon intensity
CO_2 emissions / GDP

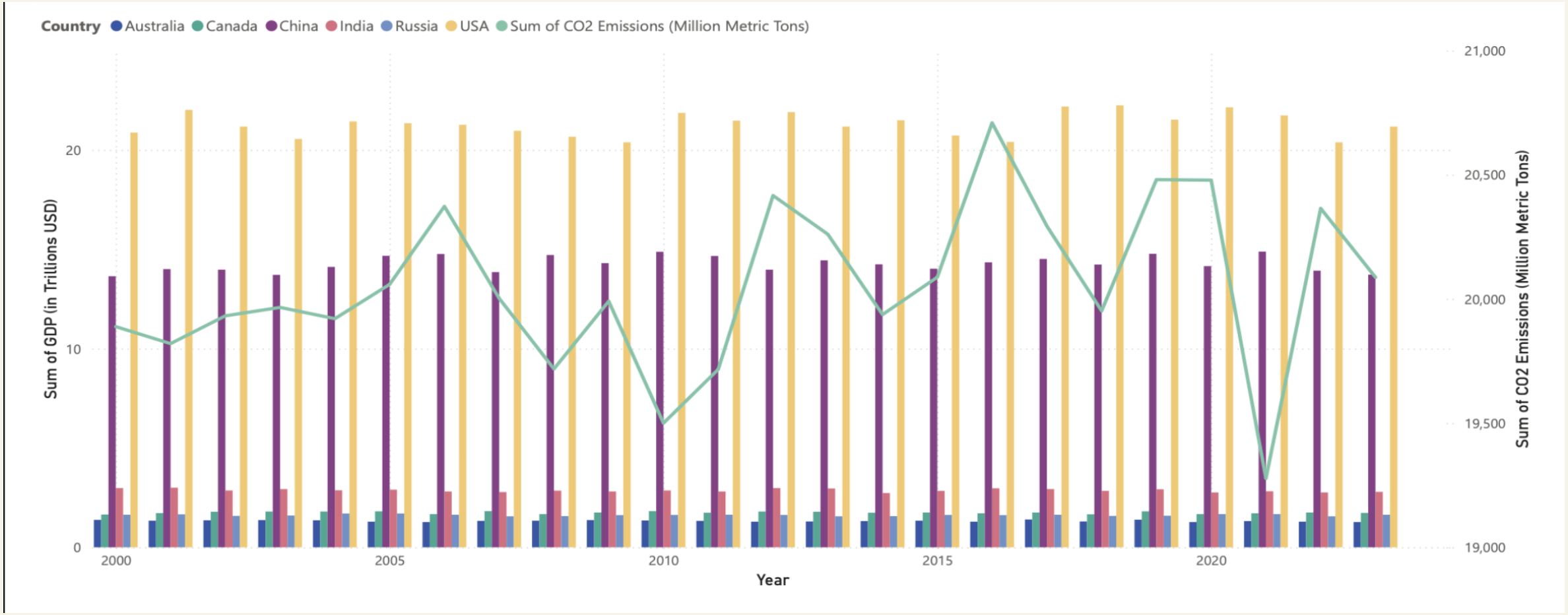
Emissions per person
CO_2 emissions / population

Resource impact questions

How does industrial activity relate to energy use?

Does economic expansion coincide with changes in land use?

Sum of GDP and CO2 Emissions by Year and Country



This chart shows overall project of GDP across the six economies for 24yrs alongside their combined CO2 emissions.

The colored bars represent each country's GDP, The green line represents total CO2 emissions across the countries.

Within this dataset, the United States consistently shows the highest GDP with China slowly catching up. The emissions line fluctuates considerably, with a noticeable decline around 2021 which right after the pandemic before rising again. While the other dip in 2010 was an inflation after the great recession.

Predictive objectives

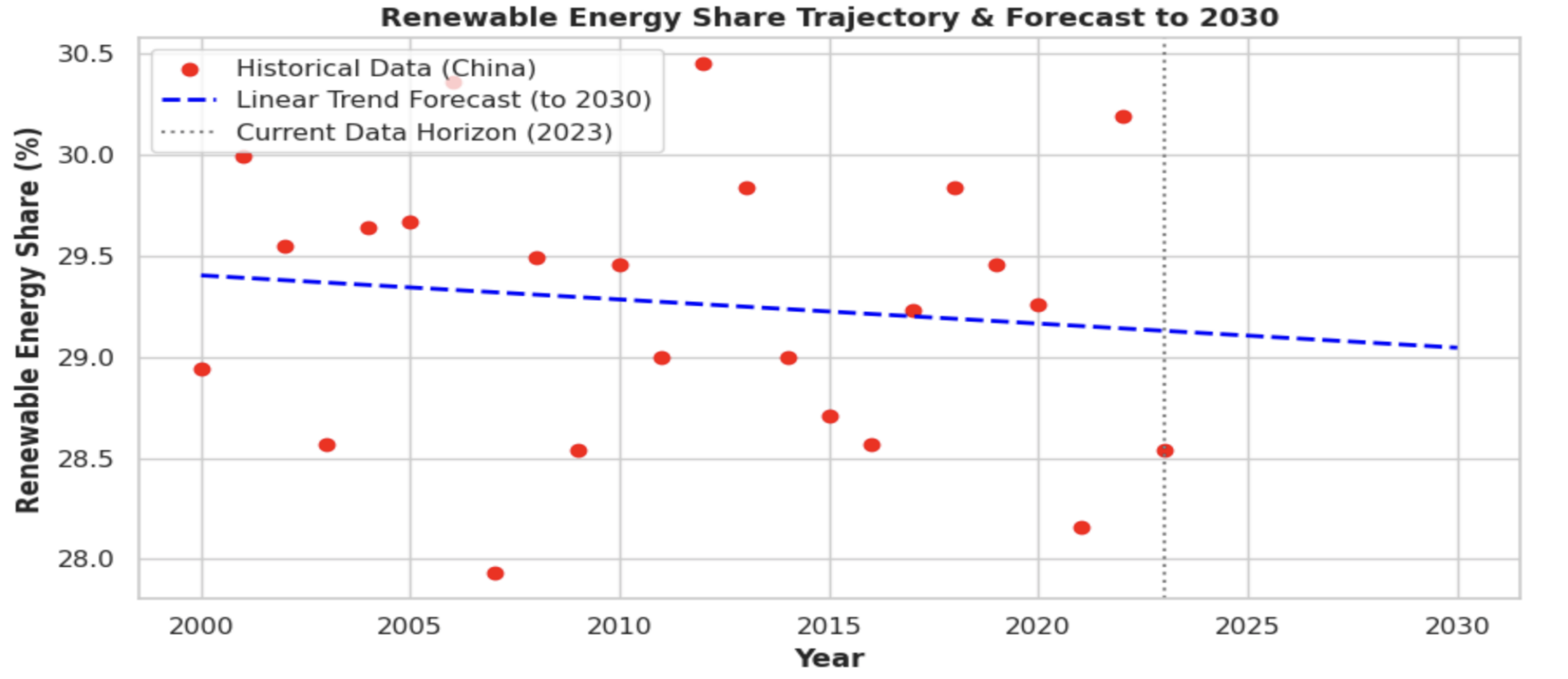
Environmental Kuznets Curve

Explore whether the emissions relationship changes as GDP per capita rises.



Predictive objectives

2030 Renewable share forecast
Proposed linear regression using historical trends.



A trend projection line chart applying linear forecasting to historical adoption rates, predicting future renewable energy share (%) through 2030 for key global economies.
Applying time-series linear trend forecasting reveals that without policy acceleration, renewable energy adoption in major industrial economies will grow at an incremental pace.
For instance, China's renewable share remains projected to hover near 28% to 30% by 2030 under historical trajectory baselines.
Historical linear adoption rates alone are insufficient to meet net-zero transition targets by 2030, highlighting the necessity for aggressive policy interventions and green capital investments.

Recent developments beyond 2023

2026 reports and announcements, as of 31 October 2026

India 13 Oct 2026	US$72bn transition investment in 2025. Coal still supplies most electricity.
California 23 Sep 2026	Announced a move toward linking its carbon market with Washington State.
Australia & Canada 25 Jun 2026	Joined the UK in pledging cooperation on clean electricity and supply chains.
Global electricity 21 Apr 2026	Renewables supplied 33.8% of electricity in 2025, overtaking coal.
Global emissions 20 Apr 2026	In 2025, GDP grew 3.1% and energy-related CO2 rose 0.4%: relative decoupling.

Clean-energy progress remains uneven, and global emissions continue to rise.

Conclusion

Greener growth progresses, but challenges remain

The project reports signs of decoupling and rising renewable adoption. Industrial energy demand remains a challenge.

Recommended priorities

Stronger policy support and clean energy investment, backed by validated country-level evidence.



References

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4. Ember Global Electricity Review 2026, https://ember-energy.org/latest-insights/global-electricity-review-2026/

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